



Alma Mater Studiorum – Università di Bologna

Department of Statistics, Economics and Business

Final Report

Forecasting Flight Departure Delays

Course: Forecasting and Predictive Analytics

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Chapter 1

Problem Statement and Objective

In recent times, air travel has become an increasingly popular mode of transportation. Thousands of flights happen every day all around the globe, requiring loads of infrastructure and tight scheduling. Air travel is by nature more complicated than other modes of transportation. It demands lots of resources from ground staff, to air traffic control, to airline staff, to complicated machinery working in tandem. That makes handling scheduling, and fluctuations in volume from day to day a challenge. As members of the Air Travel division of the US Department of Transportation, it is our job to ensure safe and smooth air travel in the United States of America. One of our biggest challenges is responding appropriately to flight delays across the US that tend to vary on a daily basis. Flight delays are logistical nightmares both from an airport operations point of view and from a commercial point of view for airlines. While we are always working to avoid delays in the first place, they are bound to happen. We believe that now, efficiency can be improved the most by anticipating and dealing swiftly with those delays to ensure that passengers have the best experience possible. Our division is seeking to build an hourly forecast for mean departure delays across major airports in the US in order to predict fluctuations in flight delays for the next day of air travel. By disseminating this forecast a day in advance to both the airport operations managers and the airline agencies in major airports, we could dramatically improve our ability to appropriately allocate resources on a given day and will thus be able to lessen the inconvenience of flight delays for passengers in the United States. Among other things, airports will be able to adjust staffing in areas such as security, air traffic control, and gate management, and more importantly will be ready to combat any potential congestion. Additionally, airlines will be more equipped to adjust their customer service staff as well as proactively alert their customers to potential delays and problems that they foresee, thus improving the overall travel experience.

Chapter 2

Data

We used a flight delay dataset obtained from Kaggle, containing 484,551 records of flights between January 1st, 2019, and June 30th, 2019. Key features include the date (Date), carrier delay (CarrierDelay), departure time (DepTime), arrival time (ArrTime), arrival delay (ArrDelay), and departure delay (DepDelay).

As a first step, we aggregated the data into hourly intervals for each day, computing the average departure delay for all flights within each one-hour slot. Upon visualizing the distribution of flights across the 24-hour day, we observed that the intervals from 3:00–4:00AM and 4:00–5:00AM had over 40% of days without any flights. Consequently, we removed these

two intervals from our analysis, defining a 22-hour "business day" ranging from midnight to 3:00AM and from 5:00AM to 11:59PM.

A few other hourly intervals still had occasional missing values due to the absence of flights on some days. To address this, we applied the `mice` package with predictive mean matching to impute the missing hourly means, ensuring a complete hourly time series suitable for subsequent time series analysis.

Following data preprocessing, we performed an analysis of the hourly distribution of delays by calculating the 25th and 75th percentiles for each hour. This quantile analysis highlighted the variability of average departure delays throughout the day, showing increased dispersion during peak hours. Such insights allowed us to better characterize the intra-day dynamics of flight delays.

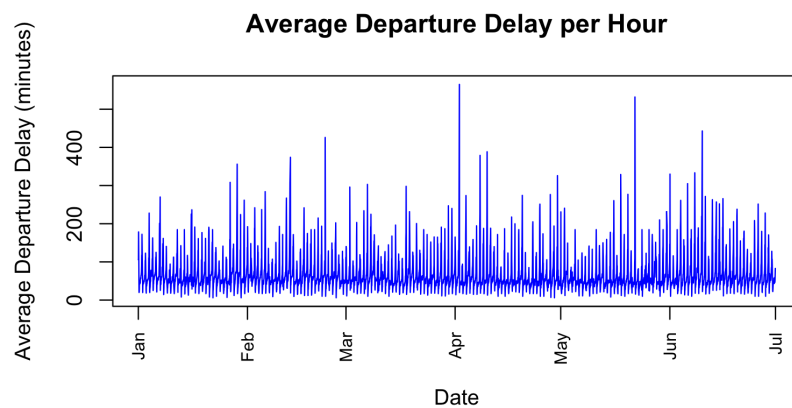


Figure 2.1: Average Departure Delay Time Series

The resulting hourly time series visualization (Figure 2.1) suggested a strong daily seasonality. Specifically, delays tended to spike in the early morning hours and then decrease as the day progressed. However, no clear weekly pattern was detected.

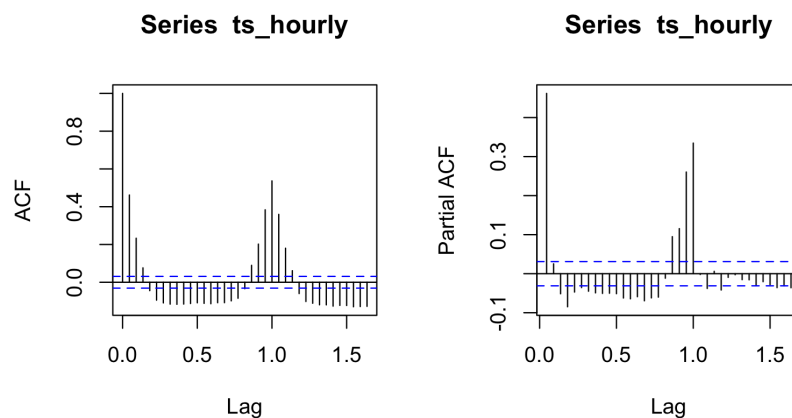


Figure 2.2: ACF and PACF of the Time Series

To further investigate the structure of the time series, we examined the autocorrelation

function (ACF) and partial autocorrelation function (PACF) plots (Figure 2.2). The ACF exhibited a wavy pattern with persistent correlations, indicating nonstationarity, while the PACF showed a similar but less pronounced structure. We also performed a seasonal-trend decomposition using the `stl` function to isolate trend, seasonal, and remainder components.

Because the trend component did not reveal a clearly deterministic structure, we tested for the presence of a stochastic trend using the Augmented Dickey-Fuller (ADF) test. The test yielded a statistic of -23.299 , significantly lower than the critical value, leading us to reject the null hypothesis and conclude that the time series does not have a stochastic trend.

Given the absence of a stochastic trend, we employed the Canova-Hansen test, which unlike the OCSB test, does not require a stochastic trend to assess seasonal stability. The test rejected the null hypothesis at the 0.05 significance level, suggesting that seasonal effects evolve over time. Additionally, the `nsdiffs` function indicated that one order of seasonal differencing is required for the series.

Chapter 3

Methodology

With the understanding that our time series is without a trend and has a seasonality that requires order one of differencing, as discussed in Chapter 2, we identified several suitable models for our forecasting problem. First, we decided to try exponential smoothing given its ability to work regardless of the presence of a seasonality. Next, given the lack of a stochastic trend, we identified a couple different forms of linear regression as options for models. To try to capture the seasonality in our time series, we also looked towards the SARIMA models. Finally, to try and account for the changes in variance across our time series, we looked to the GARCH models.

3.1 Exponential Smoothing

We started our analysis by fitting a **Simple Exponential Smoothing (SES)** model primarily as a baseline benchmark to evaluate the performance of more complex exponential smoothing models.

Given the clear presence of a seasonality in our time series, we anticipated that the **Holt-Winters method** would outperform SES by explicitly modeling the seasonal component. Holt-Winters extends SES by including both level and seasonal components, with seasonal effects modeled additively or multiplicatively. The multiplicative form is particularly appropriate when seasonal fluctuations vary proportionally with the level of the series, while additive seasonality assumes a roughly constant seasonal amplitude.

Our choice between additive and multiplicative seasonality was informed by the **Canova-Hansen test**, which indicated evolving seasonal patterns over time, suggesting multiplicative seasonality as a more appropriate choice. Nonetheless, we fitted both additive and multiplicative versions to compare their performance.

Using the `ets` function in R, we trained three models on the training set (the first 140 days of hourly observations, i.e., $22 \times 140 = 3080$ points):

1. A **Simple Exponential Smoothing model** (model = “ANN”), with additive errors (A), no trend (N), and no seasonality (N).
2. A **Holt-Winters additive seasonality model** (model = “ANA”), with additive errors (A), no trend (N), and additive seasonality (A).
3. An **automatically selected ETS model** (model = “MNM”), where the function selected multiplicative errors (M), no trend (N), and multiplicative seasonality (M) as optimal.

3.1.1 Optimal exponential model: ETS (M,N,M)

The Holt-Winters multiplicative seasonal model can be expressed as:

$$\begin{cases} \text{Level:} & \ell_t = \alpha \frac{y_t}{s_{t-m}} + (1 - \alpha)\ell_{t-1}, \\ \text{Seasonality:} & s_t = \gamma \frac{y_t}{\ell_t} + (1 - \gamma)s_{t-m}, \\ \text{Forecast:} & \hat{y}_{t+h} = \ell_t s_{t-m+h}, \end{cases} \quad (3.1)$$

where m is the seasonal period (here, daily seasonality corresponds to $m = 22$).

Our residual diagnostics confirmed that the automatically selected model (multiplicative ETS) yielded the best fit, exhibiting the lowest AIC as well as residuals with minimal autocorrelation and randomness close to white noise. Visualizations of fitted values on the training data and plots of residuals and their autocorrelation functions (ACF) further supported this conclusion, with the multiplicative model showing the most adequate fit and well-behaved residuals.

3.2 Harmonic Regression with SARMA Residual Correction

A second approach adopted to model the seasonality in the series was to fit a **Linear Regression model with Fourier (trigonometric) terms**, designed to capture the **deterministic seasonal components**. This method leverages the periodicity of the data to construct a flexible representation of cyclical patterns.

3.2.1 Harmonic Regression

We generated a grid of models with increasing numbers of harmonics (up to order 15), and used the **Akaike Information Criterion (AIC)** to select the optimal number. The selected model included:

- **11 harmonics for daily seasonality**, corresponding to the 22 hourly observations per day (i.e., $\frac{22}{2}$ harmonics),
- **1 harmonic for weekly seasonality**, with period 154 (i.e., 22×7 observations per week).

The model formula can be summarized as:

$$\begin{aligned} \text{MeanDelay}_t = & \beta_0 + \beta_1 t + \sum_{k=1}^{11} \left[\alpha_k \sin\left(\frac{2\pi kt}{22}\right) + \gamma_k \cos\left(\frac{2\pi kt}{22}\right) \right] \\ & + \delta_1 \sin\left(\frac{2\pi t}{154}\right) + \delta_2 \cos\left(\frac{2\pi t}{154}\right) + \varepsilon_t \end{aligned} \quad (3.2)$$

3.2.2 Residual Diagnostics and SARMA Correction

Despite the good fit in terms of explained variance, the **residual analysis** (ACF, PACF, Ljung-Box test) revealed persistent **autocorrelation**, particularly at seasonal lags. This indicated the presence of **stochastic seasonality** and dynamic dependencies not fully captured by the deterministic harmonics.

To model this residual structure, we fitted several **Seasonal ARMA (SARMA)** models on the regression residuals. The best-performing model (based on AIC and residual diagnostics) was selected as:

$$\text{SARMA}(4, 0, 1)(2, 0, 0)_{22} \quad (3.3)$$

This model captures both short-term autocorrelation and stochastic seasonal effects.

3.3 GARCH applied to SARMAX residuals

After estimating a regression model with harmonic terms and correcting for autocorrelation using a SARMA model, the resulting residuals were analyzed to investigate the presence of heteroskedasticity.

- **Q-Q Plot:** The plot of the residuals shows clear deviations from the reference line, particularly in the tails. This suggests that the residuals are not normally distributed.
- **ACF of Squared Residuals:** The ACF of squared residuals shows a significant increase at lag 1, indicating autocorrelation in variance. This confirms the presence of conditional heteroskedasticity, which strongly motivates the use of a GARCH model.
- **Histogram and Density Estimate:** The histogram with the kernel density estimate reveals a sharply peaked distribution centered around zero with fat tails, consistent with a non-Gaussian structure.

These findings justify the specification of a GARCH(1,1) model, where the mean equation is set to zero (as it has already been modeled using the LM + SARIMA approach) and it is

assumed that the error distribution follows a skewed normal (`snorm`) distribution to account for the observed asymmetry in the residuals.

Model Specification:

$$\varepsilon_t \sim \text{GARCH}(1, 1), \quad \varepsilon_t \sim \text{snorm}(0, \sigma_t)$$

where σ_t is the conditional variance.

Comparing the ACF plots of the squared residuals before and after applying the GARCH model (Figure 3.1), we observe that after GARCH(1,1), no significant autocorrelation remains, and all lags fall within the confidence limits.

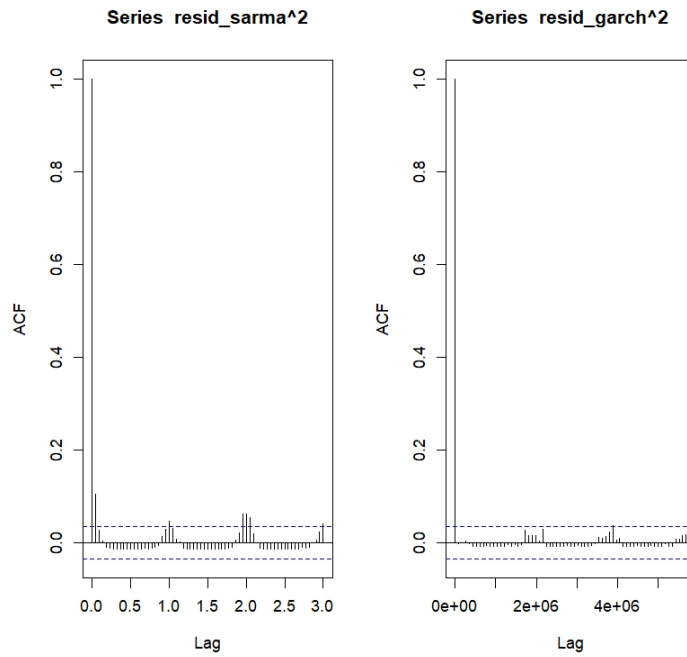


Figure 3.1: Comparison of squared residuals

This is exactly the behavior expected from a well-specified GARCH model: conditional volatility has been adequately captured, and no residual autocorrelation in the variance persists. In summary, there is no evidence of the remaining autocorrelation in the squared residuals, indicating that the GARCH(1,1) model has effectively removed the conditional heteroskedasticity.

3.4 SARIMA

After trying the Harm Regression plus SARMA approach, we decided to see how a SARIMA would perform on its own. The equation for a SARIMA is as follows:

$$\phi_P(L^s)\varphi_P(L)\nabla^d\nabla_s^D Y_t = \theta_Q(L^s)\theta_q(L)\epsilon_t \quad (3.4)$$

where

$$\nabla^d = (1 - L)^d \quad (3.5)$$

is the regular differencing operator,

$$\nabla_s^D = (1 - L^s)^D \quad (3.6)$$

is the seasonal differencing operator,

$$\varphi_p(L) \text{ and } \theta_q(L^s) \quad (3.7)$$

are regular AR and MA polynomials, and

$$\Phi_P(L^s) \text{ and } \Theta_Q(L^s) \quad (3.8)$$

are seasonal AR and MA polynomials.

Given the lack of a trend, and the optimal order of seasonal differencing of 1, we were able to fix the d and D parameters to 0 and 1 respectively. That left us to experiment with the p , q , P , and Q parameters. Based on the ACF and PACF functions, we identified three potential SARIMA models with parameters $(3,0,1)(1,0,0)$, $(4,0,1)(2,1,0)$, and $(4,0,1)(2,1,1)$. We used the the AIC metric to determine the parameters which worked best, and found that the SARIMA(4,0,1)(2,1,1) had the lowest AIC value, making it the best options.

3.5 Linear Regression with Covariates

As a predictor for departure delay, we selected the variable carrier delay. The theoretical foundation for this choice is that carrier delay is likely to influence departure delay, since the carrier is responsible for ensuring that aircraft reach their departure airports on time. As a result, any delays caused by the carrier can significantly affect the airport's ability to perform timely take-offs.

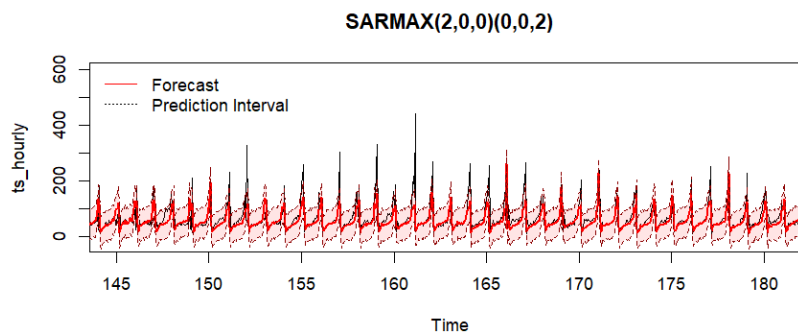


Figure 3.2: SARMAX with Carrier Delay Covariate

We began by observing that the carrier delay time series shows a pattern very similar to that of departure delay. We further analyzed the series to investigate the presence of a trend, particularly a stochastic one. We applied the Augmented Dickey-Fuller (ADF) test, which led to a rejection of the null hypothesis, allowing us to conclude that the carrier delay series does not have a stochastic trend. We then moved on to the modeling phase considering

two linear models : one based on harmonic functions and another using seasonal dummy variables. The dummy model included 22 variables to capture the daily seasonality of the data. These two approaches were selected because most of the variation in departure delay is driven by seasonal patterns. Next, we included carrier delay and a time trend as predictors in both models: the trend variable (Time) turned out to be significant only in the model with harmonic components. Based on the AIC, we selected the harmonic model as the best one, and used it for point forecasting and prediction interval. Looking at the forecast plot and residual diagnostics, the model already provided a fairly good fit, although some improvement was still possible. To enhance the model, we fitted a SARMA model to the residuals of the linear regression. After testing various specifications, we selected a SARMAX(2,0,0)(0,0,2)[22] model. As before, we generated point forecasts and prediction intervals using bootstrapping. The fit improved slightly.

Finally, to account for time-varying volatility, we fitted a GARCH(1,1) model to the residuals of the LM+SARMAX model. However, the residuals of the GARCH model still showed a slight autocorrelation, confirmed by the Box-Ljung test.

Chapter 4

Results

Using a rolling forecasting approach with a sliding window of size 22×161 (corresponding to approximately 161 days of training data, assuming 22 hourly observations per day), we generated iterative forecasts with horizon $h = 22$, i.e., one day ahead with 22 hourly predictions. The window was slid over the time series for 419 steps, retraining the models at each iteration to simulate a realistic forecasting scenario.

From Table 4.1, it emerges that the SARIMA model achieves the best metrics across the board. It has the best unconditional coverage (0.96) and the lowest Winkler Score (207.99). Instead the Diebold-Mariano test shows a positive test statistic when the SARIMA is put as model 2 in comparison with both the LM+SARMA+GARCH model and with the ETS model indicating that the SARIMA model produces statistically significantly more accurate forecasts compared to both ETS and LM+SARMA+GARCH. The ETS model demonstrates the poorest performance in terms of the Winkler Score, suggesting that its predictive intervals are insufficiently informative. In general, the SARIMA model represents the best compromise between point forecast accuracy and predictive interval quality under this rolling window evaluation framework, as we can see in Figure 4.1.

Model	Uncond. Coverage	MSE	Winkler Score	DM Test
LM+SARIMA+GARCH	0.8615	3602.621	400.84	LM+SARIMA+GARCH vs ETS: $t_n = -3.240$
ETS	0.8994	5244.581	700.69	ETS vs SARIMA: $t_n = 8.994$
SARIMA	0.9567	981.526	207.99	LM+SARIMA+GARCH vs SARIMA: $t_n = 10.358$

Table 4.1: Comparison between LM+SARMA+GARCH, ETS, and SARIMA models

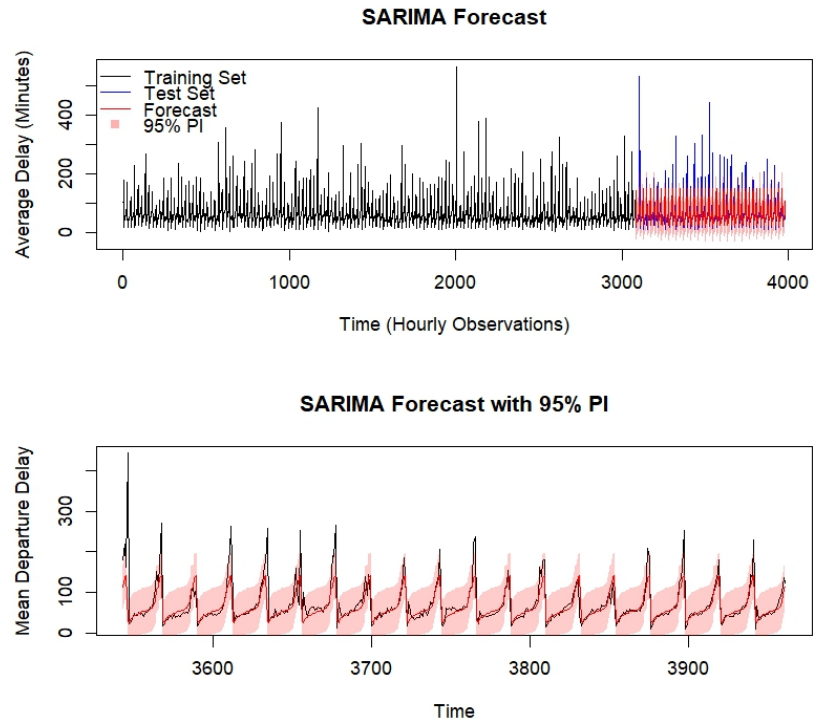


Figure 4.1: SARIMA Forecast

Chapter 5

Discussion

We have been able to create a decent forecast of mean flight departure delays using a SARIMA model. This will help both airport operations personnel and airlines in preparing for fluctuations in the next day's departure delay, thereby improving the air travel experience for all. However, there is still lots of improvement that can be made to our model. First and foremost, we were limited in the time frame of our data. Only having six months of data meant that we were missing potentially important seasonal changes that may occur to flight delays in the second half of the year. Furthermore, our data dates back to 2019. Plenty has changed in the last six years with air travel, and therefore our model is potentially not well suited for current use. It would be beneficial to have a full dataset for the past year, so that all months are included, and the data is more current. In addition, we decided to forecast a mean departure delay across major airports in the United States. However, the United States is a huge country with potentially different trends and seasonalities in different geographic areas. It may be helpful to create separate forecasts for each airport rather than lump them all into a so-called national mean. This way each airport, which also have different procedures and practices, can get better “personalized” forecasts. Finally, further analyzing models with other covariates and performing rolling-window forecasts on these models could potentially produce even more accurate results.